

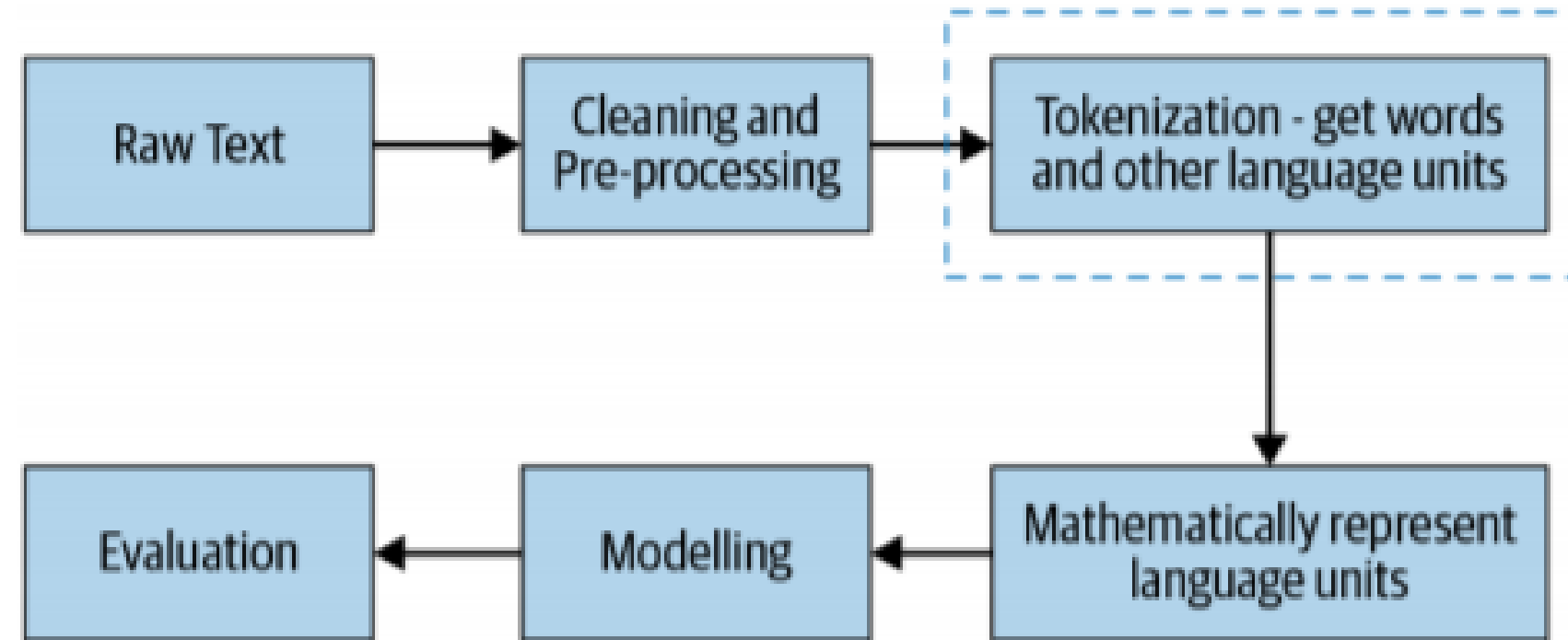
Applied NLP

DATA 641

Lecture 5 — Distributed representations

*Practical Natural Language Processing: A Comprehensive Guide to Building Real-World NLP Systems 1st Edition, Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana, O' Reilly and Natural Language Processing in Action Understanding, analyzing, and generating text with Python, Hobson Lane, Cole Howard, Hannes Hapke, ISBN 9781617294631

Text Representation



- Typical approaches of text representation
- **Distributed representations**

Why Word Embeddings?

All representation schemes we have discussed so far, we notice three fundamental drawbacks

- They are discrete representations since they treat language units as atomic units
- The feature units are sparse and high dimensional representations. In addition high dimensionality makes things computationally inefficient
- They cannot handle out-of-vocabulary (OOV) words (e.g. fastText can)

Distributed Representations

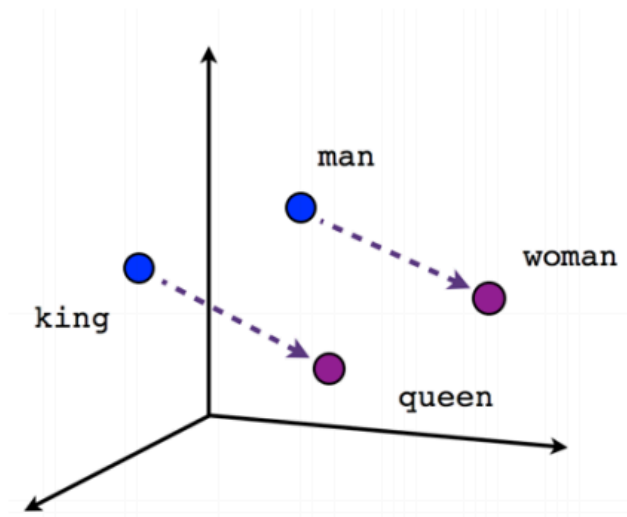
Before we look into particular methods, we need to understand some key terms

- **Distributional representation:** Refers to representation schemes that are obtained based on distribution of words from the context in which the words appear. Use high dimensional vectors to represent words. These vectors can be obtained from a co-occurrence matrix or they can be one-hot vectors. e.g. Take the sentence `I like deep learning. I like NLP. I enjoy flying`

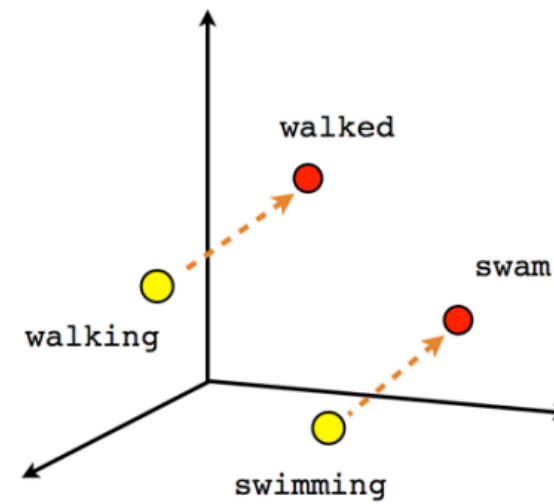
$$X = \begin{matrix} & \begin{matrix} I & like & enjoy & deep & learning & NLP & flying & . \end{matrix} \\ \begin{matrix} I \\ like \\ enjoy \\ deep \\ learning \\ NLP \\ flying \\ . \end{matrix} & \begin{bmatrix} 0 & 2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

- **Distributed representation:** Refers to representation schemes that significantly compress the dimensionality of the high dimensional vectors resulting into vectors that are compact (low dimensional) and dense (hardly any zeros).
- **Embeddings:** For the set of words in a corpus, embedding is a mapping between vector space coming from distributional representation to a vector space coming from distributed representation.

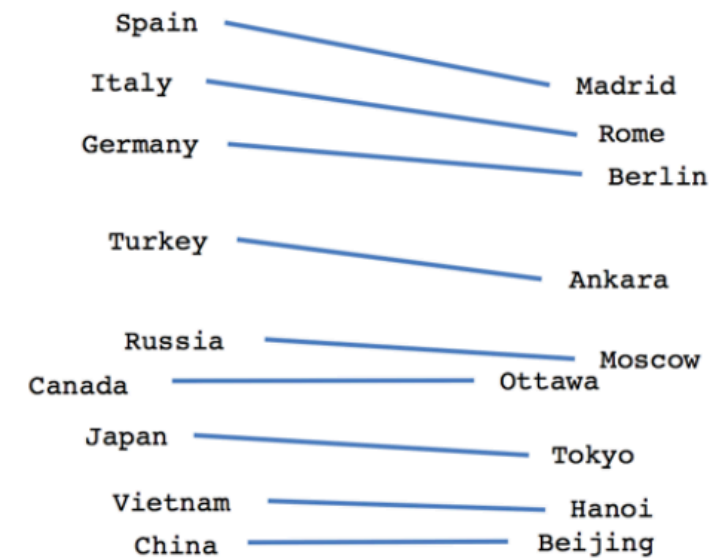
Capture "Distributional Word Similarities"



Male-Female



Verb tense



Country-Capital

Mikolov et al. proposed "Word2vec" to capture word analogy relationships: $\text{King} - \text{Man} + \text{Woman} = \text{Queen}$